UNIVERSITY OF DELAWARE · DEPARTMENT OF COMPUTER & INFORMATION SCIENCES

DAS — Initiative for Dependable Agentic Systems

Agentic systems that safely and reliably automate complex scientific and operational work

Dong Dai

Co-Director

Yixiang Deng

Co-Director

Weisong Shi

Senior Advisor

Xing Gao

Faculty Member

Team — Execution Evidence

Dong Dai
Associate Professor · Co-Director

IOAgent — IPDPS'25

STELLAR — SC'25

Agentic optimization for HPC storage and I/O workflows.

Yixiang Deng
Assistant Professor · Co-Director

Nature Communications

Cell Reports

npj Digital Medicine

AI-for-science for discovery and digital health.

Weisong Shi
Department Chair · Senior Advisor

SENIOR ADVISOR

Edge / Physical AI Strategy

Advises strategy, industry engagement, and embodied-agent directions.

Xing Gao
Assistant Professor · Faculty Member

MCP security — DSN'26

USENIX Sec'24 · CCS/S&P'24

Security for agent protocols, cloud, and supply chains.

DAS Mission - Three Pillars

Student Training
Courses · capstones · graduate research

Industry Collaboration
Consulting · TSC · pilot projects

Research Collaboration
Agent systems × domain science

Research Areas

DONG DAI

Agent Systems for System Optimization

Agents that diagnose, tune, and explain complex computing systems.

Representative work

IOAgent — IPDPS'25
STELLAR — SC'25

YIXIANG DENG

Agent × Science and Health

Scientific AI for biological discovery, digital health, and closed-loop decision support.

Representative work

AI pancreas — bioRxiv
Glucose prediction — npj Digital Med

XING GAO

Security for LLM & Agent Systems

Security analysis for agent protocols, cloud infrastructure, and software supply chains.

Representative work

MCP security — DSN'26
GPU cache attacks — USENIX Sec'24

WEISONG SHI

MENTOR

Expansion Area: Physical Agents

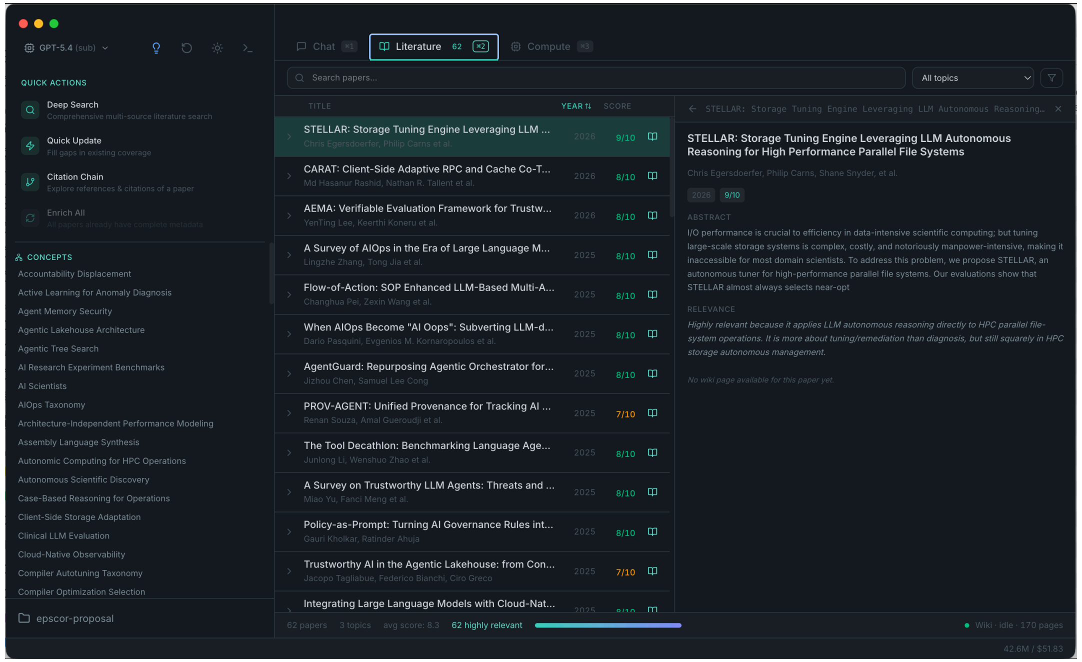
Open area for collaborators working on agents connected to instruments, robots, edge devices, or vehicles.

Status

Advisor-guided expansion
Seeking affiliates & pilots

Example Product — Research Copilot

Agentic Systems Designed for Scientific Research



Research Copilot · accountable research coworkers · daidong.github.io/PiPilot

ACCOUNTABLE RESEARCH AGENTS

Literature · data analysis · writing · project memory

Provenance / Audit
Trace every action and source.

Local Knowledge
Paper wiki and memory stay local.

Local + Cloud Models
Cloud models now; local models when needed.

How We Work with Industry

Collaboration can start as a service contract and deepen into funded research.

①	②	③	④
TSC / Consulting	**Pilot Research**	**Gift / Project Funding**	**Strategic Partnership**
Contracted expert help through UD's Technology Service Center.	A 3–6 month project around one concrete workflow and deliverable.	Support students and exploratory topics without over-scoping too early.	Shared roadmap, repeat projects, student pipeline, and joint outputs.

Start small → de-risk a pilot → fund a focused project → build a durable research channel.

TSC: sites.udel.edu/technology-service-center

What We Are Looking For

Help us choose the first dependable-agent problems worth building.

?

Founding Advisor

name / organization

TECHNICAL JUDGMENT

doors · use cases · validation

Pilot Problems

one real workflow · one clear success criterion

Flexible Support

gift · project · student support

The first advisor slots should pressure-test the roadmap and open the right conversations.

LET'S BUILD SOMETHING TOGETHER

DAS — Initiative for Dependable Agentic Systems

Dong Dai · dai@udel.edu
Yixiang Deng · yixiangd@udel.edu

Department of Computer & Information Sciences
University of Delaware

Scan to visit our site

🌐 daidong.github.io/agentic-center | 📦 Research Copilot | 🐙 GitHub